

RAK14007 WisBlock Interface Module Datasheet

Overview

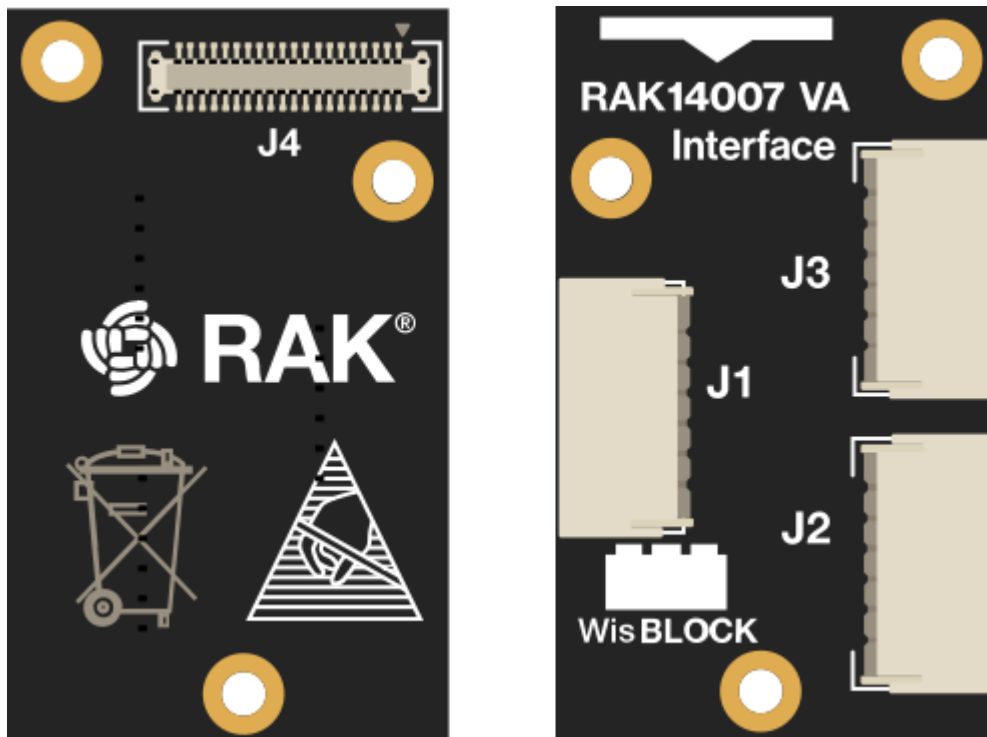


Figure 1: RAK14007 WisBlock Interface Module

Description

RAK14007 is an Interface module for connecting other WisBlock modules (like RAK14013) to the WisBlock system. It has one standard WisBlock IO connector and one connector for other WisBlock modules. With RAK14007 and a cable, some of the WisBlock modules can work out of the WisBlock Base board.

Features

- A separate joystick PCB module [RAK14013](#)  is required
- One (1) available connector for RAK14013 WisBlock Joystick Module.

- Module size: 15 × 25 mm

Specifications

Overview

Mounting

The RAK14007 module can be mounted on the IO slot of the WisBlock Base board. **Figure 2** shows the mounting mechanism of the RAK14007 on a WisBlock Base board, such as the RAK5005-O.

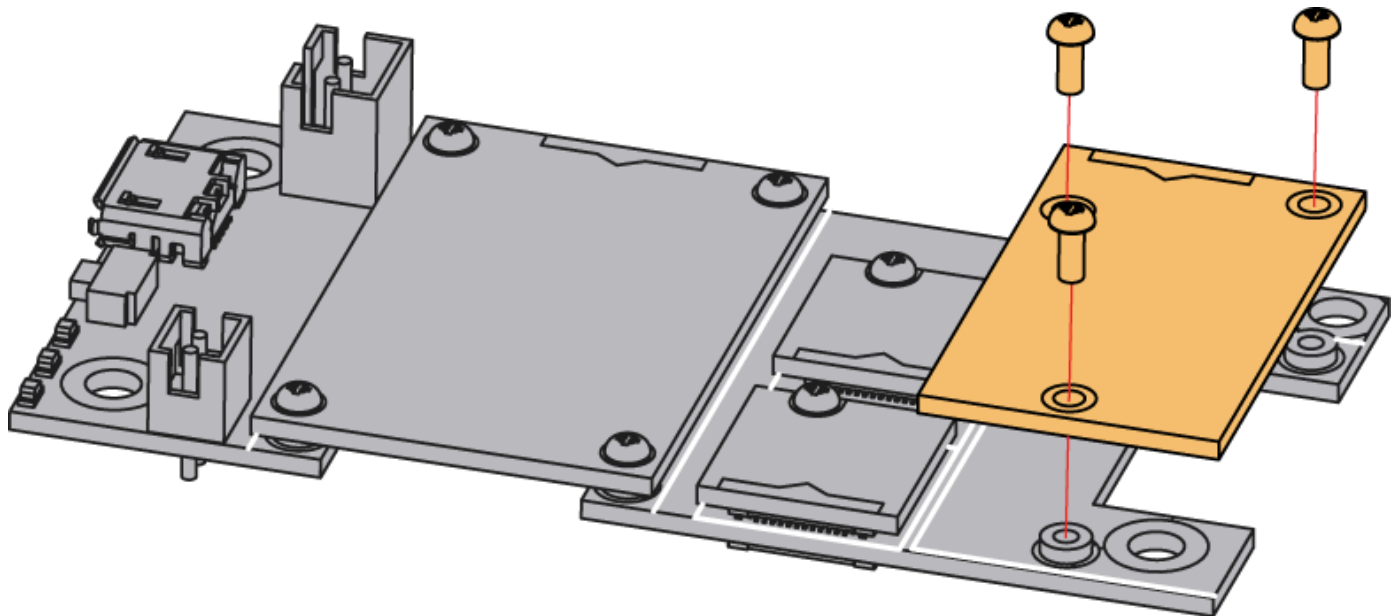


Figure 2: RAK14007 mounting mechanism on a WisBlock Base module

Hardware

The hardware specification is categorized into three (3) parts. It discusses the pinouts and their corresponding functions and diagrams of the module. Also, it shows the mechanical characteristics of the RAK14007 WisBlock Interface Module.

Pin Definition

The RAK14007 module has a 40-pin WisConnector that is compatible with the WisBlock Base IO Slot. The pin order of the connector and the pinout definition is shown in **Figure 3**.

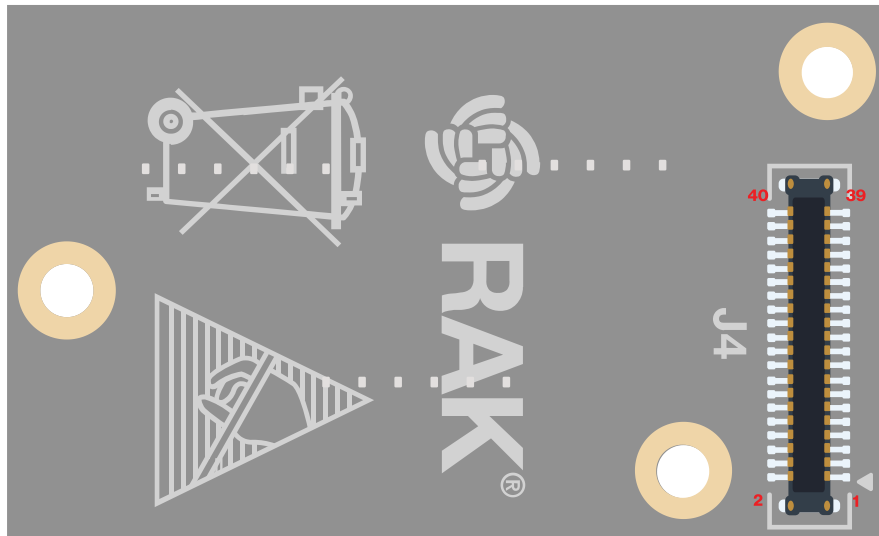


Figure 3: RAK14007 Pinout Schematic

VBAT	2	1	VBAT
GND	4	3	GND
3V3_S	6	5	NC
NC	8	7	NC
NC	10	9	NC
NC	12	11	NC
NC	14	13	NC
NC	16	15	NC
NC	18	17	NC
I2C_SCL	20	19	I2C_SDA
AIN1	22	21	NC
NC	24	23	NC
NC	26	25	NC
NC	28	27	NC
NC	30	29	NC
IO4	32	31	IO3
NC	34	33	NC
NC	36	35	NC
IO6	38	37	IO5
GND	40	39	GND

NOTE

- 3V3_S, VBAT, I2C related pin, AIN1, IO3, IO4, IO5, and GND are connected to WisConnector.

Mechanical Characteristics

Board Dimensions

Figure 4 shows the mechanical dimensions of the RAK14007 Module.

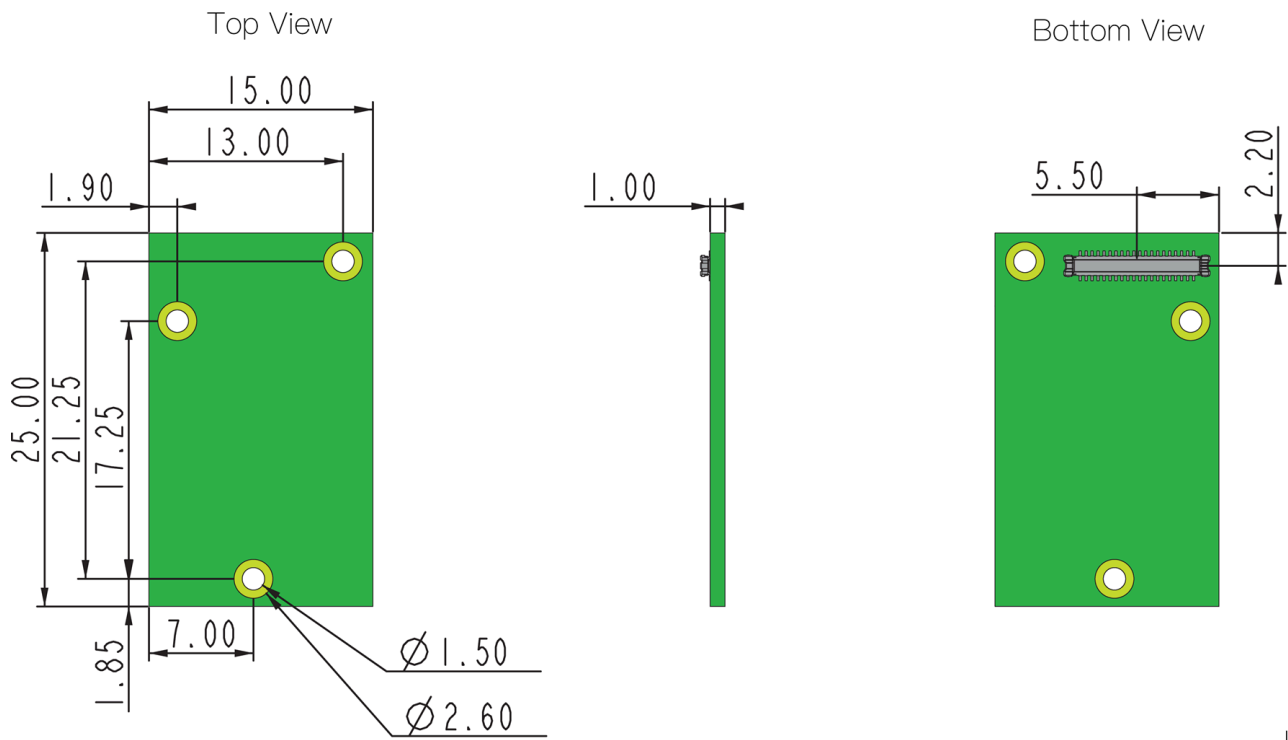


Figure 4: RAK14007 mechanical dimensions

WisConnector PCB Layout

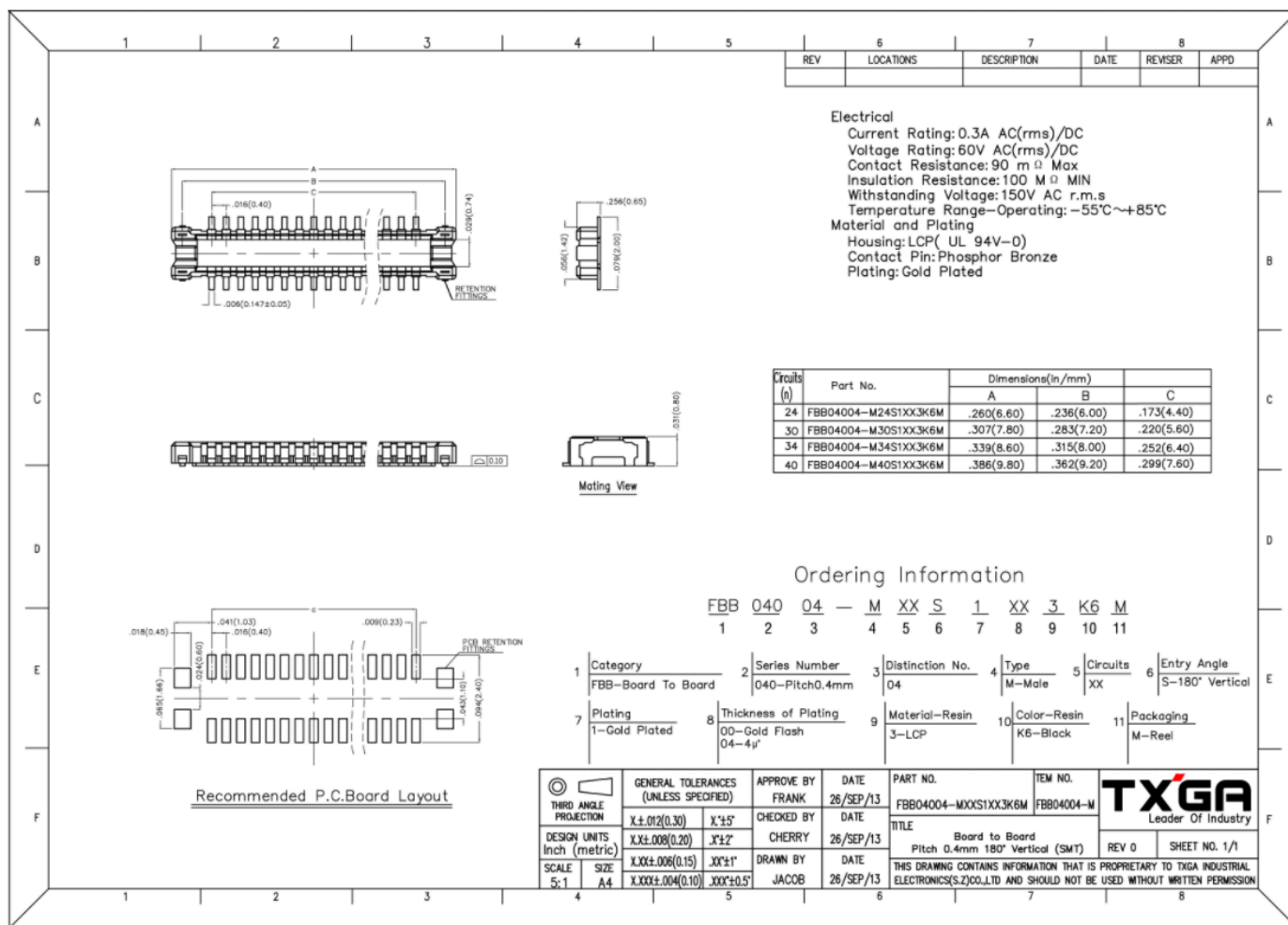


Figure 5: WisConnector PCB footprint and recommendations

Schematic Diagram

Figure 6 shows the schematic of the RAK14007 module.

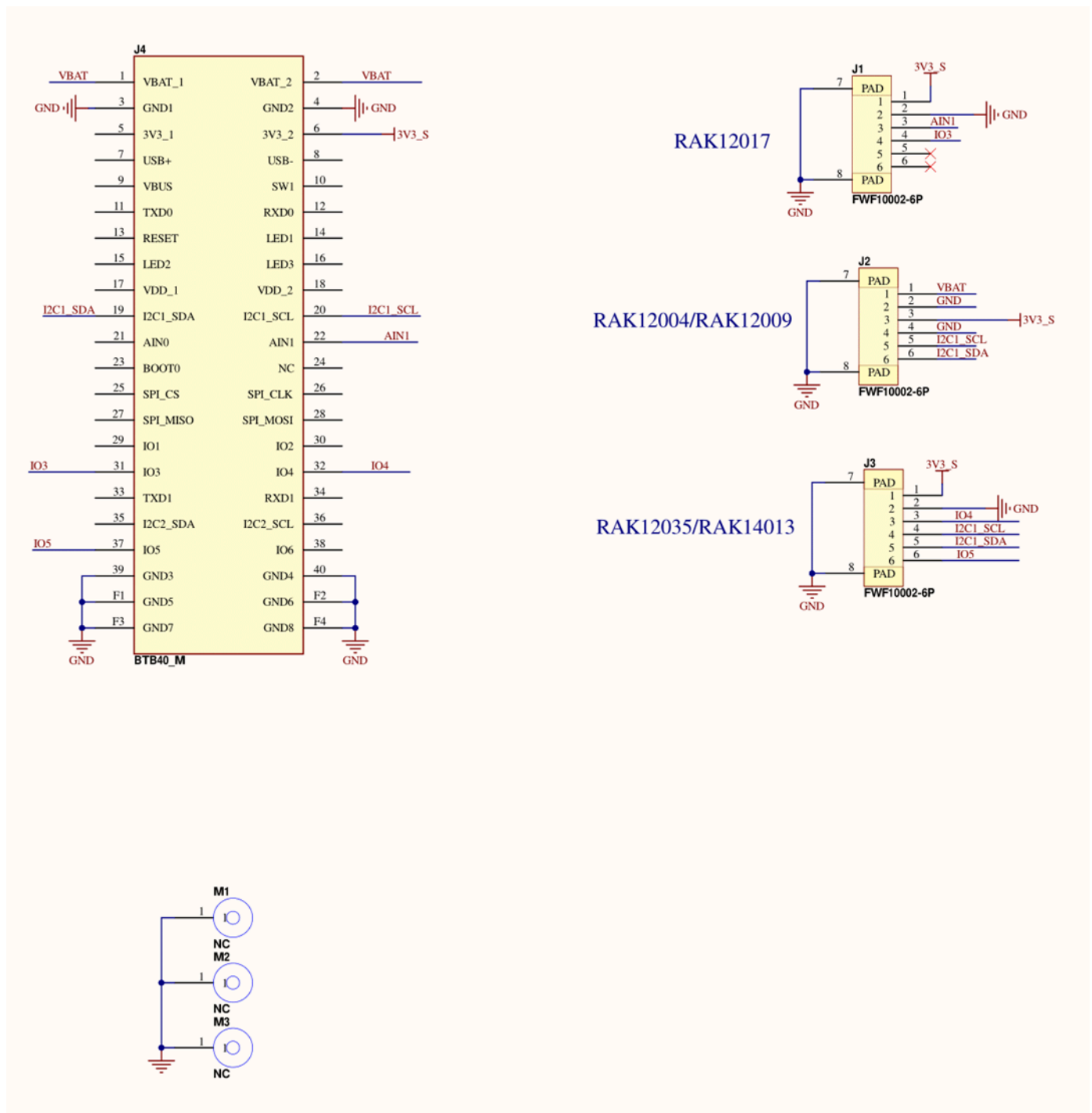


Figure 6: RAK14007 WisBlock Interface Module schematics

NOTE

The following modules are now available to access RAK14007. There may be other modules in the future.

- RAK12017 can connect to J1.

- RAK12004 and RAK12009 can connect to J2.
- RAK12035 and RAK14013 can connect to J3.